Gluu-EMA--Solo.io-Gluu-Gluu

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MCP Security Interoperability Test (SIT) Dec 2026

EMA Interoperability Testing Log

File name: `Gluu-EMA--Solo.io-Gluu-Gluu`
Tester: Gluu (Mohibuzzaman Zico, zico@gluu.org). All times UTC. Both Gluu servers ran with DEBUG logging for the test.

When Performed

Started: 2026-10-13 16:51:16 UTC
Ended: 2026-10-13 16:52:27 UTC
Where performed: online

Participants

Role	Company and Product	Representative	Email
MCP Client	Solo.io: Solo Enterprise for agentgateway, kagent Enterprise	Nick Gamb	nick.gamb@solo.io
OAuth AS	Gluu: Gluu Flex / Janssen (ledger-demo.gluu.info)	Mohibuzzaman Zico	zico@gluu.org
OpenID Provider	Gluu: Gluu Flex / Janssen (sterling-idp.gluu.info)	Mohibuzzaman Zico	zico@gluu.org

Test Report

MCP Client Operation

Gluu holds both the OAuth AS and the OpenID Provider roles in this test, so the same person verified both sides. All marks are provisional (☑). The OAuth AS and OpenID Provider sections are removed.

Title	MCP Client (Solo.io)
Valid ID-JAG	☑ (ledger-demo validated the ID-JAG the client presented: logs 3 and 4)
resource support	☑ (the client sent `resource` in the token exchange: log 1; the issued ID-JAG carries it: log 5)
scope support	☑ (the client sent `scope=research:read` in both requests: logs 1 and 3; the ID-JAG carries it: log 5)

Test Logs, Screenshots and Videos

Representative exchange at 16:51:16 to 16:51:17 UTC. The same sequence repeats three more times at 16:51:17 and once at 16:52:26 UTC (shown last).

Log 1. sterling-idp (OpenID Provider), 16:51:16. RFC 8693 token exchange (private_key_jwt client authentication)

```
2026-10-13 16:51:16,912 DEBUG [qtp1783047508-16] 83d16253-6f4d-4ef2-96a0-6b2e21688f8e
[as.server.token.ws.rs.TokenRestWebServiceImpl] (TokenRestWebServiceImpl.java:171) - Attempting to request access
token: grantType = urn:ietf:params:oauth:grant-type:token-exchange, code = <redacted>, redirectUri = null, username
= null, refreshToken = null, clientId = fcffc488-b086-4a69-a6c5-47d15c4f00c4, ExtraParams =
{subject_token_type=[urn:ietf:params:oauth:token-type:id_token], audience=[https://ledger-demo.gluu.info],
client_assertion_type=[urn:ietf:params:oauth:client-assertion-type:jwt-bearer],
grant_type=[urn:ietf:params:oauth:grant-type:token-exchange], resource=[https://mcp.ledgerline.lab/mcp],
scope=[research:read], client_assertion=<redacted>, subject_token=<redacted>,
requested_token_type=[urn:ietf:params:oauth:token-type:id-jag], client_id=[fcffc488-b086-4a69-a6c5-47d15c4f00c4]},
isSecure = true, codeVerifier = <redacted>, ticket = null, authorizationDetails = null
```

Log 2. sterling-idp, ID-JAG issued with audience ledger-demo

```
2026-10-13 16:51:16,916 DEBUG [qtp1783047508-16] 83d16253-6f4d-4ef2-96a0-6b2e21688f8e
[as.server.token.ws.rs.IdJagService] (IdJagService.java:118) - Issued ID-JAG for client: fcffc488-b086-4a69-
a6c5-47d15c4f00c4, audience: https://ledger-demo.gluu.info
```

Log 3. ledger-demo (OAuth AS), 16:51:17. jwt-bearer grant (private_key_jwt client authentication)

```
2026-10-13 16:51:17,061 DEBUG [qtp1176735295-17] 9f2c97ba-a5b3-47a4-827e-d0437e53b609
[as.server.token.ws.rs.TokenRestWebServiceImpl] (TokenRestWebServiceImpl.java:170) - Attempting to request access
token: grantType = urn:ietf:params:oauth:grant-type:jwt-bearer, code = <redacted>, redirectUri = null, username =
null, refreshToken = null, clientId = fcffc488-b086-4a69-a6c5-47d15c4f00c4, ExtraParams =
{client_assertion_type=[urn:ietf:params:oauth:client-assertion-type:jwt-bearer],
grant_type=[urn:ietf:params:oauth:grant-type:jwt-bearer], scope=[research:read], assertion=<redacted>,
client_assertion=<redacted>, client_id=[fcffc488-b086-4a69-a6c5-47d15c4f00c4]}, isSecure = true, codeVerifier =
<redacted>, ticket = null, authorizationDetails = null
```

Log 4. ledger-demo, ID-JAG validated, user resolved, scope granted

```
2026-10-13 16:51:17,063 DEBUG [qtp1176735295-17] 9f2c97ba-a5b3-47a4-827e-d0437e53b609
[as.server.token.ws.rs.IdJagValidatorService] (IdJagValidatorService.java:88) - Validating ID-JAG for client:
fcffc488-b086-4a69-a6c5-47d15c4f00c4
2026-10-13 16:51:17,068 DEBUG [qtp1176735295-17] 9f2c97ba-a5b3-47a4-827e-d0437e53b609
[as.server.token.ws.rs.IdJagValidatorService] (IdJagValidatorService.java:231) - Resolved user by 'email', user: bob
2026-10-13 16:51:17,068 DEBUG [qtp1176735295-17] 9f2c97ba-a5b3-47a4-827e-d0437e53b609
[as.server.token.ws.rs.IdJagValidatorService] (IdJagValidatorService.java:100) - ID-JAG validated successfully for
client: fcffc488-b086-4a69-a6c5-47d15c4f00c4
2026-10-13 16:51:17,070 DEBUG [qtp1176735295-17] 9f2c97ba-a5b3-47a4-827e-d0437e53b609
[jans.as.server.model.authorize.ScopeChecker] (ScopeChecker.java:91) - Granted scopes: [research:read]
```

Log 5. Decoded ID-JAG and ledger-demo's response, from the MCP client's gateway log (provided by Solo.io, signature removed)

```
ID-JAG issued by Gluu (sterling-idp.gluu.info) at 2026-10-13T16:52:25.993Z for the account-info tool call
(2026-10-13T16:52:27.034Z, trace 1efeb728a02b7b6c1c197ea4df2b4618)
header: {"kid":"connect_506b48d1-79c3-45b1-8463-eddda3a5c41d_sig_rs256","typ":"oauth-id-jag+jwt","alg":"RS256"}
payload:
acr: fido2
amr: ["70"]
aud: https://ledger-demo.gluu.info
auth_time: 1791910247
client_id: fcffc488-b086-4a69-a6c5-47d15c4f00c4
email: bob@sterling.lab
exp: 1791910646
iat: 1791910346
iss: https://sterling-idp.gluu.info
jti: 23eec5d8-0105-4a75-8fd3-1af7d2094de4
resource: https://mcp.ledgerline.lab/mcp
scope: research:read
sub: 9HX4wEtMFfR7ViNrDMHp1DwQto6qdkdnJhG6ZDn1Hjo
Requested in the token exchange: requested_token_type=urn:ietf:params:oauth:token-type:id-jag, audience=https://
ledger-demo.gluu.info, resource=https://mcp.ledgerline.lab/mcp, scope=research:read
Checks: typ oauth-id-jag+jwt; aud "https://ledger-demo.gluu.info"; client_id fcffc488-b086-4a69-a6c5-47d15c4f00c4;
resource https://mcp.ledgerline.lab/mcp; scope research:read; sub_id not issued (sub only); sid not issued
```

```
Gluu ledger-demo accepted the ID-JAG: POST /xaa-legs/ledgerline/as/token (grant_type=urn:ietf:params:oauth:grant-
type:jwt-bearer) -> 200 at 2026-10-13T16:52:26.133Z (137ms)
response: token_type=Bearer, issued_token_type=urn:ietf:params:oauth:token-type:access_token, scope=research:read,
expires_in=5
access token header: {"kid":"connect_261dceaa-e858-4872-b61a-6bc54aa209e8_sig_rs256","typ":"JWT","alg":"RS256"}
access token payload: {"iss":"https://ledger-demo.gluu.info","sub":"004fb040-80df-4abf-8f81-eee69263cf55","aud":
["fcffc488-b086-4a69-a6c5-47d15c4f00c4","ledgerline-research"],"client_id":"fcffc488-b086-4a69-
a6c5-47d15c4f00c4","scope":"[research:read]","exp":1791910351,"iat":1791910346}
```

Second exchange, 16:52:26 UTC

sterling-idp

```
2026-10-13 16:52:26,025 DEBUG [qtp1783047508-20] 3a386c38-cc6a-45c5-9bbb-f78e6299a094
[as.server.token.ws.rs.TokenRestWebServiceImpl] (TokenRestWebServiceImpl.java:171) - Attempting to request access
token: grantType = urn:ietf:params:oauth:grant-type:token-exchange, code = <redacted>, redirectUri = null, username
= null, refreshToken = null, clientId = fcffc488-b086-4a69-a6c5-47d15c4f00c4, ExtraParams =
{subject_token_type=[urn:ietf:params:oauth:token-type:id_token], audience=[https://ledger-demo.gluu.info],
client_assertion_type=[urn:ietf:params:oauth:client-assertion-type:jwt-bearer],
grant_type=[urn:ietf:params:oauth:grant-type:token-exchange], resource=[https://mcp.ledgerline.lab/mcp],
scope=[research:read], client_assertion=<redacted>, subject_token=<redacted>,
requested_token_type=[urn:ietf:params:oauth:token-type:id-jag], client_id=[fcffc488-b086-4a69-a6c5-47d15c4f00c4]},
isSecure = true, codeVerifier = <redacted>, ticket = null, authorizationDetails = null
2026-10-13 16:52:26,028 DEBUG [qtp1783047508-20] 3a386c38-cc6a-45c5-9bbb-f78e6299a094
[as.server.token.ws.rs.IdJagService] (IdJagService.java:118) - Issued ID-JAG for client: fcffc488-b086-4a69-
a6c5-47d15c4f00c4, audience: https://ledger-demo.gluu.info
```

ledger-demo

```
2026-10-13 16:52:26,161 DEBUG [qtp1176735295-20] 7eb0ff4b-b462-42c7-ba22-52d636eb2b3e
[as.server.token.ws.rs.TokenRestWebServiceImpl] (TokenRestWebServiceImpl.java:170) - Attempting to request access
token: grantType = urn:ietf:params:oauth:grant-type:jwt-bearer, code = <redacted>, redirectUri = null, username =
null, refreshToken = null, clientId = fcffc488-b086-4a69-a6c5-47d15c4f00c4, ExtraParams =
{client_assertion_type=[urn:ietf:params:oauth:client-assertion-type:jwt-bearer],
grant_type=[urn:ietf:params:oauth:grant-type:jwt-bearer], scope=[research:read], assertion=<redacted>,
client_assertion=<redacted>, client_id=[fcffc488-b086-4a69-a6c5-47d15c4f00c4]}, isSecure = true, codeVerifier =
<redacted>, ticket = null, authorizationDetails = null
2026-10-13 16:52:26,162 DEBUG [qtp1176735295-20] 7eb0ff4b-b462-42c7-ba22-52d636eb2b3e
[as.server.token.ws.rs.IdJagValidatorService] (IdJagValidatorService.java:88) - Validating ID-JAG for client:
fcffc488-b086-4a69-a6c5-47d15c4f00c4
2026-10-13 16:52:26,166 DEBUG [qtp1176735295-20] 7eb0ff4b-b462-42c7-ba22-52d636eb2b3e
[as.server.token.ws.rs.IdJagValidatorService] (IdJagValidatorService.java:231) - Resolved user by 'email', user: bob
2026-10-13 16:52:26,166 DEBUG [qtp1176735295-20] 7eb0ff4b-b462-42c7-ba22-52d636eb2b3e
[as.server.token.ws.rs.IdJagValidatorService] (IdJagValidatorService.java:100) - ID-JAG validated successfully for
client: fcffc488-b086-4a69-a6c5-47d15c4f00c4
2026-10-13 16:52:26,168 DEBUG [qtp1176735295-20] 7eb0ff4b-b462-42c7-ba22-52d636eb2b3e
[jans.as.server.model.authorize.ScopeChecker] (ScopeChecker.java:91) - Granted scopes: [research:read]
```

Notes and disclosures

- sterling-idp.gluu.info runs Janssen 2.5.0. ledger-demo.gluu.info runs Janssen 2.4.1 (Flex).
- Token values (client_assertion, assertion, subject_token, code, code_verifier, state, code_challenge) are replaced by `<redacted>`. Everything else is raw jans-auth.log output.
- Janssen verifies an ID-JAG signature only with the server's own keystore. For this run the signing certificate of sterling-idp was imported into the keystore of ledger-demo (under its kid) as a workaround. Fetching the trusted issuer's keys from a jwks_uri or a static JWKS is not available yet.
- `sub_id` is not issued by Gluu (the ID-JAG carries `sub` only), so that cell is blank.
- Not verified: the MCP client does not send `resource` in the jwt-bearer request to the AS, so resource support at the AS is not exercised.